Year 8 Earth and Space Sciences Revision Guide answers

Definitions:

- Rock
  The solid mineral material forming part of the surface of the earth and other similar planets, exposed on the surface or underlying the soil
- Mineral
  Substances that make up rocks each mineral has its own chemical make up.
- Crystal
  Geometrically-shaped substance made up of atoms and molecules arranges in one of seven different shapes. The elements that make up a crystal and the conditions present during the crystal's growth determine the arrangement of atoms and molecules and the shape of the crystals.
- Igneous rocks
  Rocks that form from the cooling of lava or magma as it is thrown through the air from a volcanic eruption.
- Intrusive rock
  Igneous rock that forms when magma cools below the Earth's surface.
- Extrusive rock
  Igneous rock that forms when lava cools above the Earth's surface.
- Sedimentary rock
  Rocks formed from sediments deposited by water, wind or ice. The sediments are cemented together in layers, under pressure.
- Erosion
  A type of weathering in which surface soil and rock are worn away through the action of glaciers, water, and wind
- Sediments
  Material broken down by weathering and erosion that is moved by wind or water and collects in layers.
- Fossils
  Any remains, impression, or trace of an animal or plant of a former geological age.
- Layering
  The action of arranging something in layers
- Floodplains
  An area of low-lying ground adjacent to a river, formed mainly of river sediments and subject to flooding.
- Coal
  A sedimentary rock formed from dead plants and animals that were buried before rotting completely
- Metamorphic rocks
  A rock formed from another rock that has been under great heat and/or pressure
- Parent rock
  The original rock from which something else was formed.
- Rock cycle
  An idealized cycle of processes undergone by rocks in the earth's crust, involving igneous intrusion, uplift, erosion, transportation, deposition as sedimentary rock, metamorphism, and further melting and igneous intrusion

- Mineral ore
  Rocks mined to obtain a metal or other chemical within them
- Mining
  Extraction of natural resources from the Earth
- Environmental impact statement
  Study of the possible effects of a planned project on the environment
- Rehabilitation
  The act of restoring something to its original state, like the rehabilitation of the forest that had once been cleared for use as a mine.
- Overburden
  Waste rock removed from below the topsoil. This rock is replaced when the area is restored.
- Open cut mining
  Mining that scours out soil and rocks on the surface of the land
- Underground mining
  Mining that uses shafts and tunnels to remove rock from deep below the surface

Questions:

1) For each of the three different types of rocks, describe their characteristics, how they are formed, and some examples of them.

	Igneous	Sedimentary	Metamorphic
How it is formed	The solidification of magma and lava	Weathered grains of rock or biological material.	Igneous rock or sedimentary rock which is exposed to great heat and or pressure over time.
Examples	Basalt, Obsidian, etc	Limestone, Shale, etc	Marble, Quartzite
Does it have layers?	No	Yes	Yes/can be
Is it porous?	Can be	Yes	No
Level of hardness?	Ranges	Ranges	Hard
Does it have grains or crystals?	Crystals	Grains	Both
The level of effect of	Medium	High	Low

weathering			
Diagram			

1. Write a description for each type of weathering

Weathering		
Chemical	Rain water	Rainwater is naturally slightly acidic. Minerals in rocks may react with the rainwater, causing the rock to be weathered
	Acid rain	Acid rain is more acidic than rainwater and makes chemical weathering happen more quickly.
Biological	Animals	Burrowing animals such as rabbits can burrow into a crack in a rock, making it bigger and splitting the rock.
	Plants	**Plant roots** can grow in cracks. As they grow bigger, the roots push open the cracks and make them wider and deeper. Eventually pieces of rock may fall away.
	Humans	Caused by people walking. Over time, paths in the countryside become damaged because of all the boots and shoes wearing them away.
Physical	Exfoliation	When a rock gets hot it **expands** a little, and when a rock gets cold it **contracts** a little. If a rock is heated and cooled many times, **cracks** form and **pieces** of rock **fall** away.
	Wind, rain and waves	Wind, rain and waves can all cause weathering. The wind can blow tiny grains of sand against a rock. These wear the rock away and weather it. Rain and waves can also wear away rock over long periods of time.
	Freeze-thaw process	If water gets into a crack in a rock and then **freezes**, it **expands** and pushes the crack further apart. When the ice melts later, water can get deeper into the crack. When the rock freezes again, it expands and makes the crack even bigger. This process of freezing and thawing can continue until the crack becomes so big that a piece of rock falls off.

2) What are the similarities and differences between an intrusive rock, and an extrusive rock?
Intrusive rocks and extrusive rocks are both igneous in nature. However, intrusive rocks are formed by magma (i.e., below ground) and extrusive rocks are formed by the expulsion of lava (i.e., above ground).

3) Some igneous rocks have pores in them. Are these rocks intrusive or extrusive? Why are there pores in the rock?
These rocks are extrusive. The pores in the rock originate from the gases that were present in the lava as it was cooling in the air.

4) What are the major differences between igneous rocks and sedimentary rocks?
The biggest difference is that igneous rocks are made up of crystals, whereas sedimentary rocks form by the cementing of grains.
5) Describe the process in which sedimentary rocks are formed.
Rock is weathered, resulting in the rock breaking into smaller pieces. A body of water usually carries these pieces along, until the water slows down (like at the mouth of a lake, or the sea). The sediments are then deposited on the bed. Compaction occurs; the weight of sediments on the top start to squash the sediments on the bottom. The water is then squeezed out from the bottom layer, which causes crystals to form. This is called cementation. This process keeps occurring between different layers, and over a lot of time, eventually forming sedimentary rocks.

6) Why is running water an important part of the formation of sedimentary rocks?
Water moves the sediments from one point to another, and the point at which water slows down is the area in which sedimentary rock will start to form.

7) What are the two types of materials can form sedimentary rocks?
Biological material and mineral grains.

8) How are igneous and sedimentary rocks transformed into metamorphic rocks?
Through exposure to large amounts of heat and pressure, over time.

9) What are minerals?
Minerals are chemical substances within rocks with a definite chemical composition.

10) What daughter rocks do Limestone and Sandstone form under high heat?
Marble and Quartzite
11) What daughter rock does Shale form under high pressure?
Slate

12) What are the different types of weathering?
Physical, biological and chemical

13) When walking through a park, you notice a large bed of rock has started to crack from it's underside. A large tree is nearby. What type of weathering is this? Explain how it occurs.
This is biological weathering. As the tree has grown over time, the roots have increased in size. This results in pressure building up on the underside of the rock, causing the rock to break over time.

14) A large rock face is situated in an area which, over the course of the year, experiences an average temperature of around -5 degrees in winter, and 30 degrees in summer. Explain why there are deep cracks in the rock face. HINT: What does an average temperature mean?
This could be the result of both the changes in the temperature, and freeze-thaw action.
An increase in temperature would cause rocks to expand, and to contract on cooling, exacerbating any smaller cracks/spaces in the rock.
Freeze-thaw could occur during the winter months when the temperature would range from positive to negative. If any rainfall occurs, this means that water can get into cracks; when the temperature is negative it will freeze, causing the water to expand, making the cracks larger; and then evaporate off.

15) What type of rocks are most susceptible to weathering?
Chalk, limestone

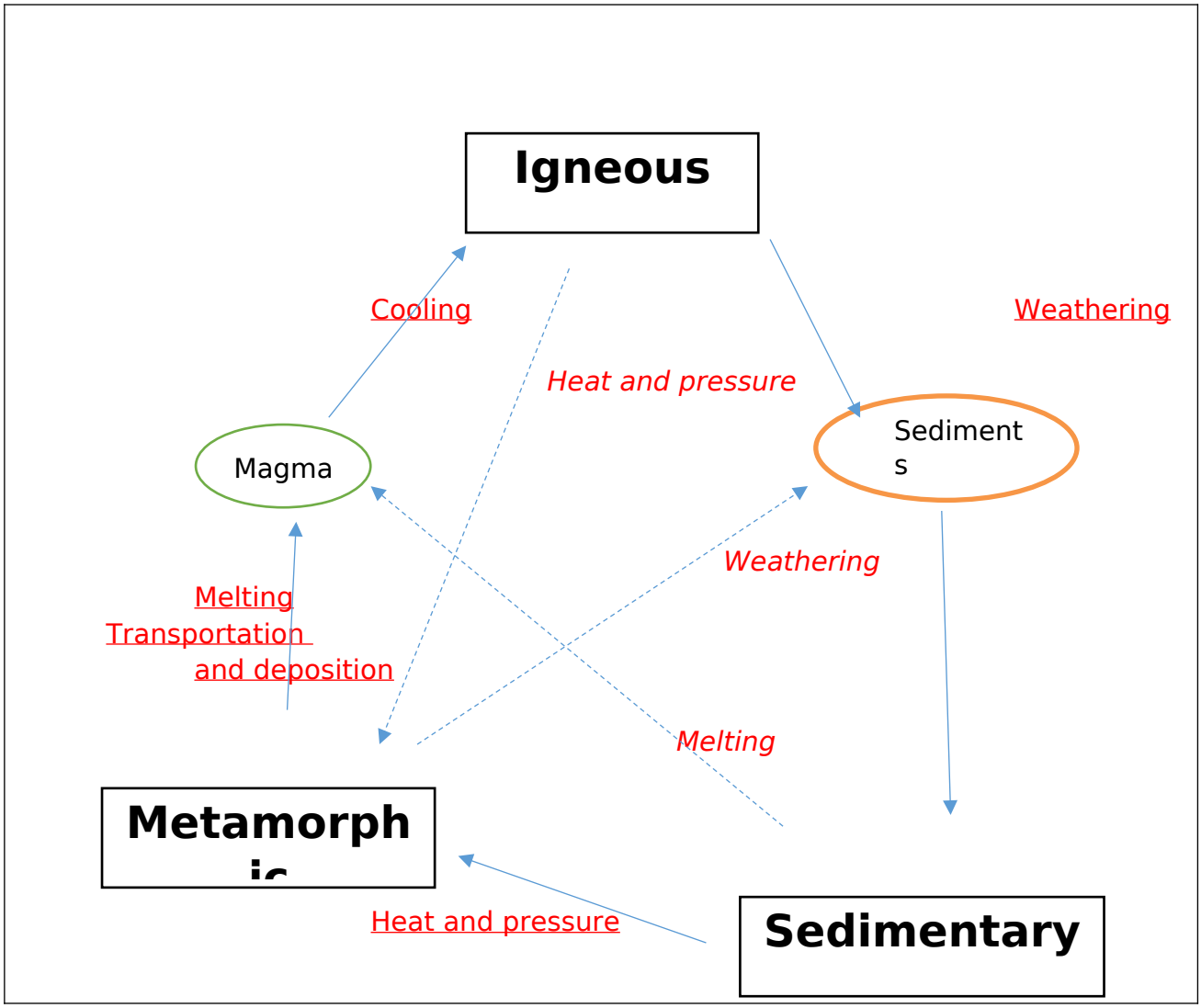
16) Explain the difference between weathering and erosion (use definitions and an example).
Weathering is the breaking down of rock. Erosion is when the particle are moved away.

17) What are the two main types of mines?
Underground, Open-cut

18) What is required for a company to start mining?
Environmental impact statement

19) How do mining companies get metal from ore?
Chemical processes

20) Draw a labelled diagram of the rock cycle. Some has been done for you.



21) Label the following examples of rock as either igneous, sedimentary or metamorphic
   a. Limestone: Sedimentary
   b. Slate: Metamorphic

c. Conglomerate: Conglomerate
d. Pumice: Igneous
e. Rock salt: Sedimentary
f. Marble: Metamorphic
g. Granite: Igneous
h. Basalt: Igneous
i. Coal: Sedimentary
j. Shale: Sedimentary